Jenkins CI/CD Pipeline

I. Manual Workflow - without using Jenkins :

Étape 1 : Créer le fichier Dockerfile

1. Créer le fichier Dockerfile :

```
arij@ubuntu26:~$ git clone https://github.com/wahid007/Jenkins-Demo.git
Cloning into 'Jenkins-Demo'...
remote: Enumerating objects: 197, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 197 (delta 2), reused 2 (delta 2), pack-reused 194 (from 2)
Receiving objects: 100% (197/197), 94.52 KiB | 1.55 MiB/s, done.
Resolving deltas: 100% (90/90), done.
arij@ubuntu26:~$ cd Jenkins-Demo
```

2. Créez un fichier Dockerfile :

```
  GNU nano 8.7.1                                                Dockerfile *
FROM eclipse-temurin:17-jre
EXPOSE 8080
WORKDIR /app
COPY target/*.jar app.jar
CMD ["java", "-jar", "app.jar"]
```

Étape 2 : Construire l'image Docker :

1. Compilez le projet avec Maven :

```
arij@ubuntu26:~/Jenkins-Demo$ mvn clean install
```

```
[INFO] Installing /home/arij/Jenkins-Demo/target/devOpsDemo-0.0.1-SNAPSHOT.jar to /home/arij/.m2/repository/com/devops/devOpsDemo/0.0
.1-SNAPSHOT/devOpsDemo-0.0.1-SNAPSHOT.jar
[INFO] Installing /home/arij/Jenkins-Demo/pom.xml to /home/arij/.m2/repository/com/devops/devOpsDemo/0.0.1-SNAPSHOT/devOpsDemo-0.0.1-
SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:35 min
[INFO] Finished at: 2026-10-13T22:09:55Z
[INFO] ------------------------------------------------------------------------
arij@ubuntu26:~/Jenkins-Demo$
```

2.Construisez l'image Docker :

```
arij@ubuntu26:~/Jenkins-Demo$ docker build -t jenkins-demo:1.0.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  24.52MB
Step 1/5 : FROM eclipse-temurin:17-jre
17-jre: Pulling from library/eclipse-temurin
a920497d91be: Pulling fs layer
109bc5b81bb6: Pulling fs layer
15bf01d33b4f: Pulling fs layer
a7267ff71f0e: Pulling fs layer
4e07a0f12b2c: Pulling fs layer
06ad70e463aa: Pulling fs layer
15bf01d33b4f: Download complete
06ad70e463aa: Download complete
a920497d91be: Download complete
12a193872bff: Download complete
05199d4393f0: Download complete
109bc5b81bb6: Download complete
a7267ff71f0e: Download complete
4e07a0f12b2c: Download complete
06ad70e463aa: Pull complete
4e07a0f12b2c: Pull complete
109bc5b81bb6: Pull complete
15bf01d33b4f: Pull complete
a920497d91be: Pull complete
a7267ff71f0e: Pull complete
Digest: sha256:207ecae0b2b104dfc6dfa763d1e9cd2041cb344800a180e24cb1a28ed533ff90
Status: Downloaded newer image for eclipse-temurin:17-jre
 ---> 207ecae0b2b1
Step 2/5 : EXPOSE 8080
 ---> Running in f465867750d9
```

Étape 3 : Pousser l'image sur Docker Hub :

a.Connecter à Docker Hub :

```
arij@ubuntu26:~/Jenkins-Demo$ docker login -u arijbelaid

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

WARNING! Your credentials are stored unencrypted in '/home/arij/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
arij@ubuntu26:~/Jenkins-Demo$
```

b. Taguer votre image :

```
arij@ubuntu26:~/Jenkins-Demo$ docker tag jenkins-demo:1.0.0 arijbelaid/jenkins-demo:1.0.0
arij@ubuntu26:~/Jenkins-Demo$
```

c. Pousser l'image :

```
arij@ubuntu26:~/Jenkins-Demo$ docker push arijbelaid/jenkins-demo:1.0.0
The push refers to repository [docker.io/arijbelaid/jenkins-demo]
06ad70e463aa: Mounted from library/eclipse-temurin
e177b1f84d0d: Pushed
a7267ff71f0e: Mounted from library/eclipse-temurin
15bf01d33b4f: Mounted from library/eclipse-temurin
a920497d91be: Mounted from library/eclipse-temurin
123f40839abd: Pushed
109bc5b81bb6: Mounted from library/eclipse-temurin
4e07a0f12b2c: Mounted from library/eclipse-temurin
1.0.0: digest: sha256:5fec756710490240ab79ee9d3e0ba35eb7b5a76d7577dda548e3c099926d0003 size: 2051
arij@ubuntu26:~/Jenkins-Demo$
```

d.créer un conteneur :

```
arij@ubuntu26:~/Jenkins-Demo$ docker run --detach --name container-jenkins --publish 2222:2222 arijbelaid/jenkins-demo:1.0.0
a15cefc9e8e51ae6e5ee082acaf09ef412ea0daee1f6d71452a81231fb7f9e28
arij@ubuntu26:~/Jenkins-Demo$
```

e.Tester :

```
arij@ubuntu26:~/Jenkins-Demo$ curl http://localhost:2222
<!DOCTYPE html>
<html>
  <head>
    <title>Spring Boot HTML Page</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        margin: 0;
        padding: 0;
      }

      h1 {
        text-align: center;
        color: #333;
        margin-top: 100px;
      }

      p {
        text-align: center;
        color: #666;
      }

      .container {
        text-align: center;
        margin-top: 20px;
      }
```

Étape 4 : Déployer sur Kubernetes (K3S/Minikube) :

a. Créer le fichier deployment.yaml et service.yaml

```
arij@ubuntu26:~$ cd ~/Jenkins-Demo
nano deployment.yaml
arij@ubuntu26:~/Jenkins-Demo$ nano service.yaml
```

b. Appliquer le Deployment

```
arij@ubuntu26:~/Jenkins-Demo$ kubectl apply -f deployment.yaml
deployment.apps/spring-app-deployment created
```

c. Vérifier les Pods

```
^Carij@ubuntu26:~/Jenkins-Demo$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
spring-app-deployment-7dbff8f6b5-57fjp   1/1     Running   0          3m55s
spring-app-deployment-7dbff8f6b5-jkgt5   1/1     Running   0          3m55s
arij@ubuntu26:~/Jenkins-Demo$
```

d. Appliquer le Service

```
arij@ubuntu26:~/Jenkins-Demo$ kubectl apply -f service.yaml
service/spring-app-service created
arij@ubuntu26:~/Jenkins-Demo$
```

e. vérifier le service de deploiement :

```
arij@ubuntu26:~/Jenkins-Demo$ kubectl get svc
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1       <none>        443/TCP          21m
spring-app-service   NodePort    10.102.25.36    <none>        2222:30263/TCP   6m2s
arij@ubuntu26:~/Jenkins-Demo$
```

f.tester l'application :

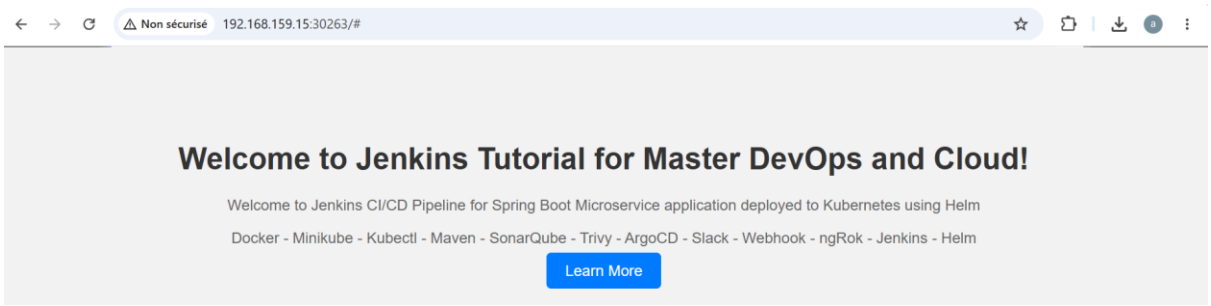
```
arij@ubuntu26:~/Jenkins-Demo$ curl http://192.168.159.15:30263
<!DOCTYPE html>
<html>
  <head>
    <title>Spring Boot HTML Page</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        margin: 0;
        padding: 0;
      }

      h1 {
        text-align: center;
        color: #333;
        margin-top: 100px;
      }

      p {
        text-align: center;
        color: #666;
      }

      .container {
        text-align: center;
        margin-top: 20px;
      }

      .btn {
        background-color: #007bff;
        color: #fff;
        padding: 10px 20px;
```

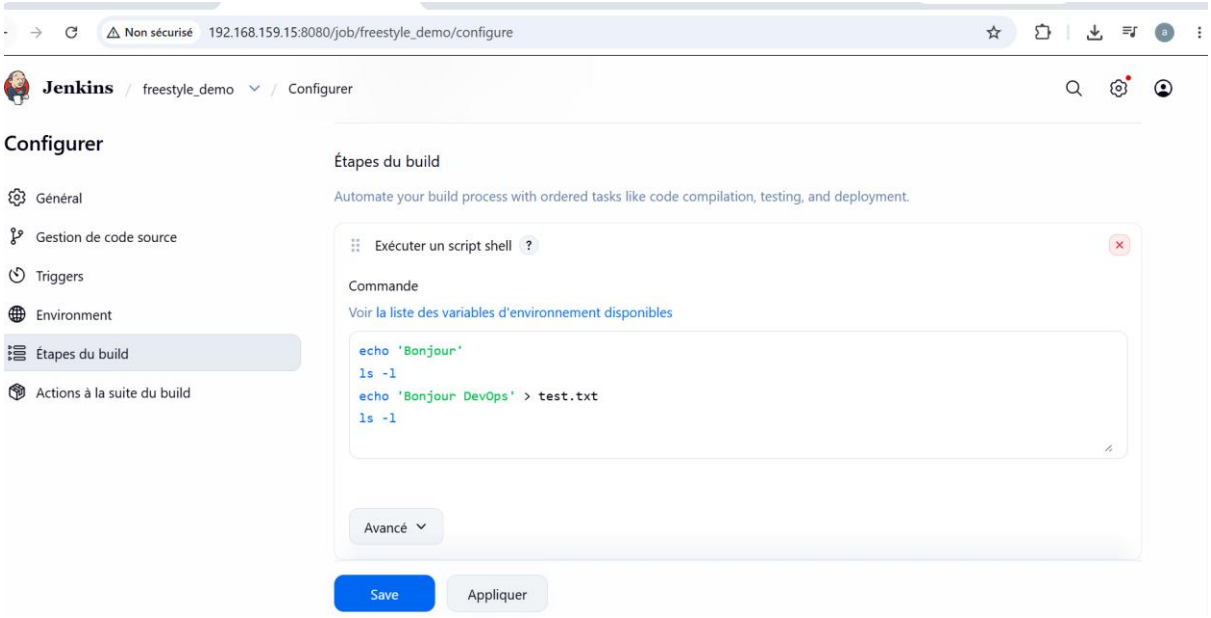


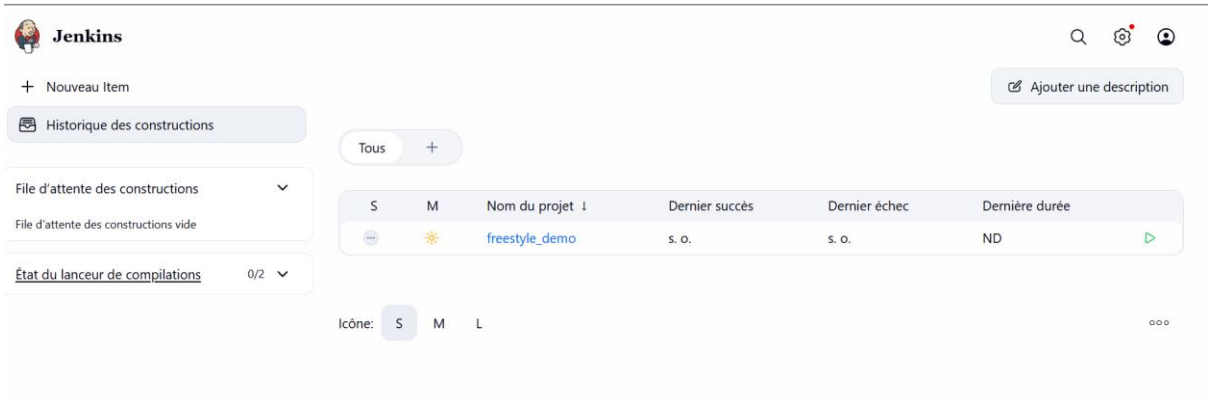
II. Jenkins Freestyle Workflow :

```
arij@ubuntu26:~/Jenkins-Demo$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
bbeeac7e3d5044738e586baa7e5257e7
arij@ubuntu26:~/Jenkins-Demo$
```

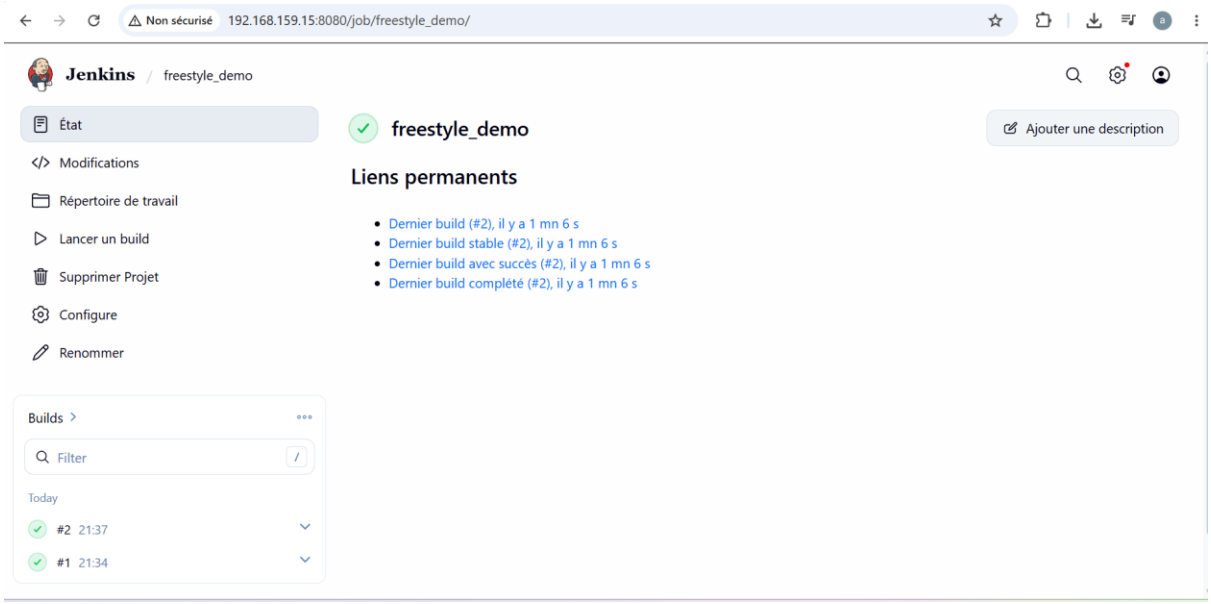
Step_1 :

1 .Créer le job freestyle_demo :

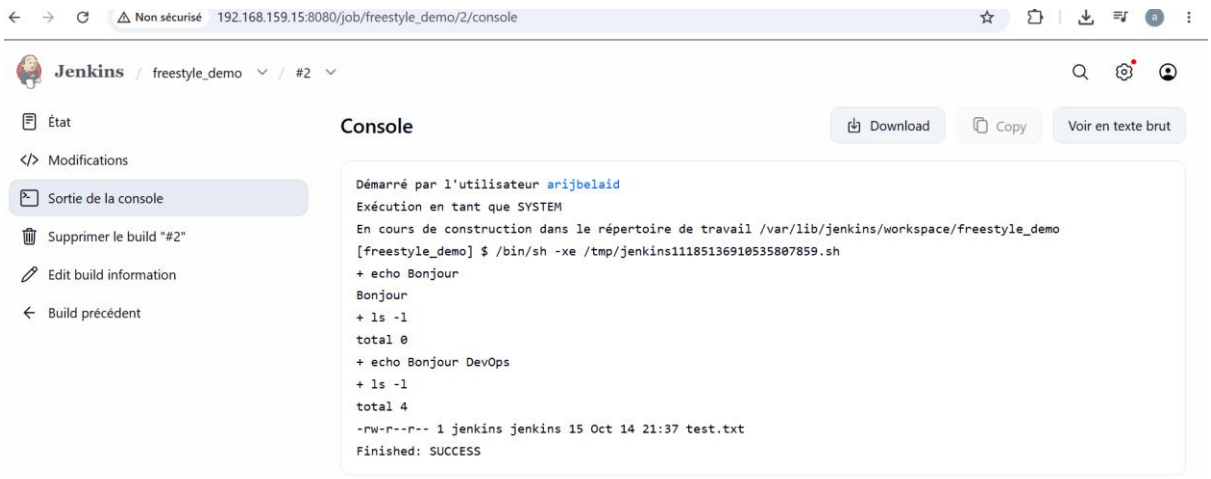




2. Lancer le build



3.Vérifier :



4.Vérification sur vm :

```
arij@master:~$ sudo ls -l /var/lib/jenkins/workspace/freestyle_demo
total 4
-rw-r--r-- 1 jenkins jenkins 15 Oct 14 21:37 test.txt
arij@master:~$ sudo cat /var/lib/jenkins/workspace/freestyle_demo/test.txt
Bonjour DevOps
arij@master:~$
```

Step_2 :

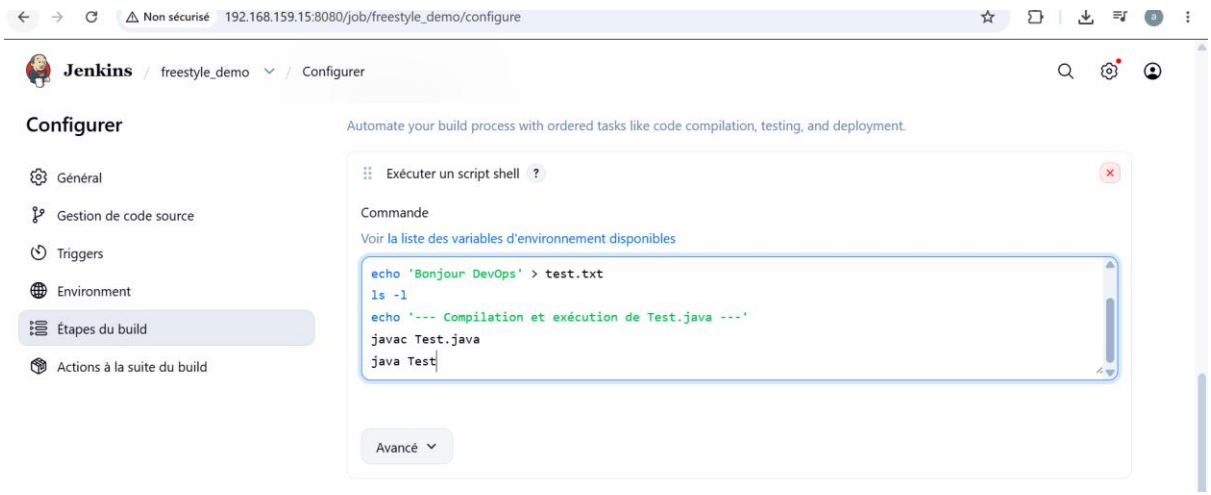
1. Créer le fichier Test.java :

```
 GNU nano 8.7.1                          /var/lib/jenkins/workspace/freestyle_demo/Test.java *
public class Test {
    public static void main(String[] args) {
        System.out.println("Hi Master DevOps");
    }
}
```

2.Donnez les droits à l'utilisateur jenkins sur le fichier :

```
arij@master:~$ sudo chown jenkins:jenkins /var/lib/jenkins/workspace/freestyle_demo/Test.java
arij@master:~$
```

3. Modifier le job Jenkins :



4. Lancer le build



Step_3 :

1.Créer le fichier script.sh et Test.java

```
 GNU nano 8.7.1                                          script.sh *
#!/bin/bash
echo 'Bonjour depuis GitHub'
ls -l
echo 'Bonjour DevOps' > test.txt
ls -l
echo '--- Compilation et exécution de Test.java ---'
javac Test.java
java Test
```

```
 GNU nano 8.7.1                                          test.java *
public class Test {
    public static void main(String[] args) {
        System.out.println("Hi Master DevOps");
    }
}
```

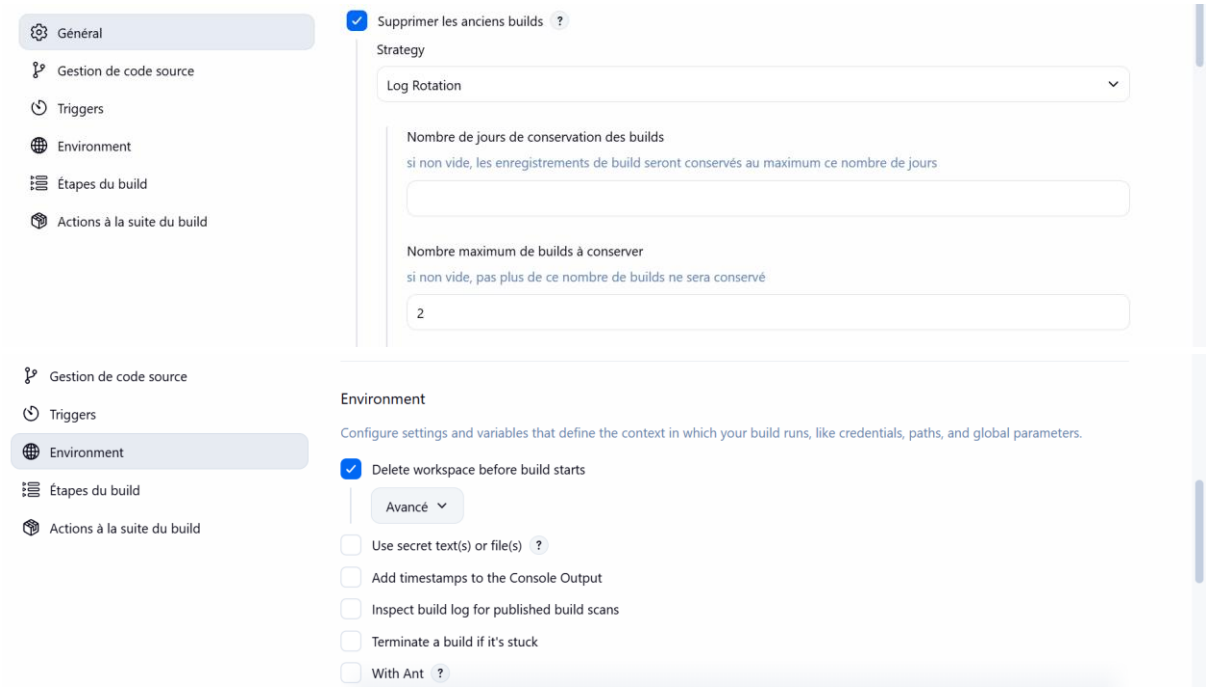
2.pusher sur github :

```
arij@master:~/jenkins-scripts$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 755 bytes | 47.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:arijbelaid/jenkins-scripts.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
arij@master:~/jenkins-scripts$
```

Step_4 :

1.Configuration du





Step_5 :

1. Ajouter jenkins au groupe docker

```
arij@master:~/jenkins-scripts$ sudo usermod -aG docker jenkins
[sudo: authenticate] Password:
```

2.Copier le kubeconfig pour jenkins

```
arij@master:~/jenkins-scripts$ sudo mkdir -p /var/lib/jenkins/.kube
arij@master:~/jenkins-scripts$ sudo cp /home/arij/.kube/config /var/lib/jenkins/.kube/config
arij@master:~/jenkins-scripts$ sudo chown -R jenkins:jenkins /var/lib/jenkins/.kube
arij@master:~/jenkins-scripts$ sudo chmod 600 /var/lib/jenkins/.kube/config
```

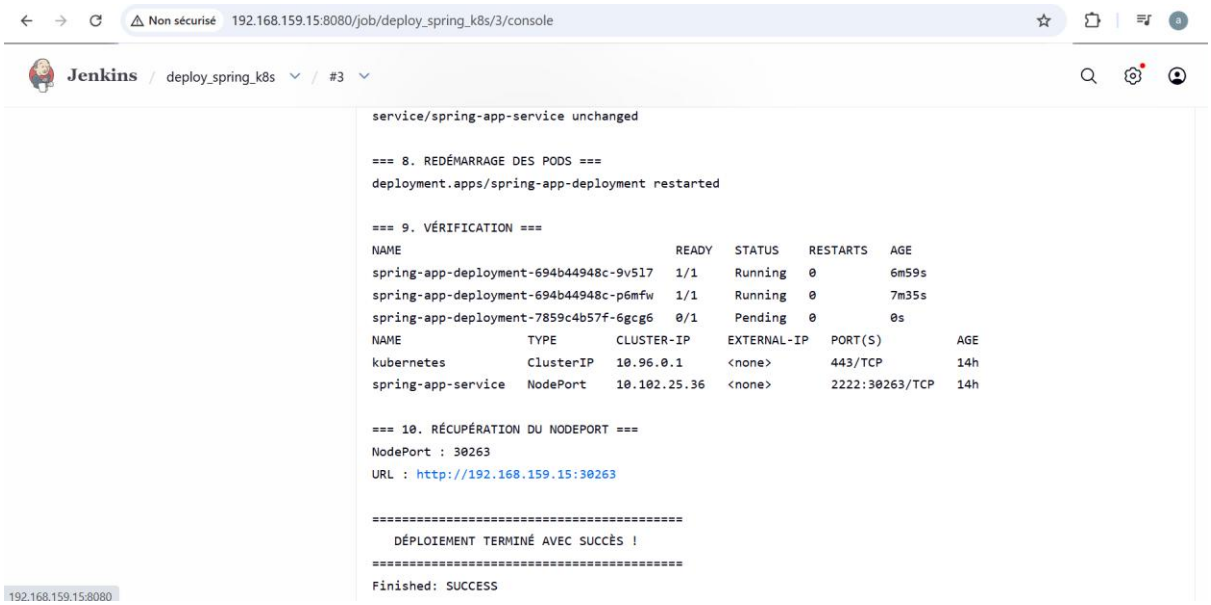
3. Vérifier les droits :

```
arij@master:~/jenkins-scripts$ sudo -u jenkins docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
arij@master:~/jenkins-scripts$ sudo -u jenkins kubectl get nodes
NAME      STATUS   ROLES           AGE   VERSION
master    Ready    control-plane   14h   v1.31.14
worker1   Ready    <none>          14h   v1.31.14
worker2   Ready    <none>          14h   v1.31.14
arij@master:~/jenkins-scripts$
```

4. Créer le nouveau job +lancer :



5.Vérification :



6.Vérification sur vm :

```
arij@master:~/jenkins-scripts$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
spring-app-deployment-7859c4b57f-6gcg6   1/1     Running   0          97s
spring-app-deployment-7859c4b57f-vwrql   1/1     Running   0          83s
arij@master:~/jenkins-scripts$
```

7.Test d’application :

